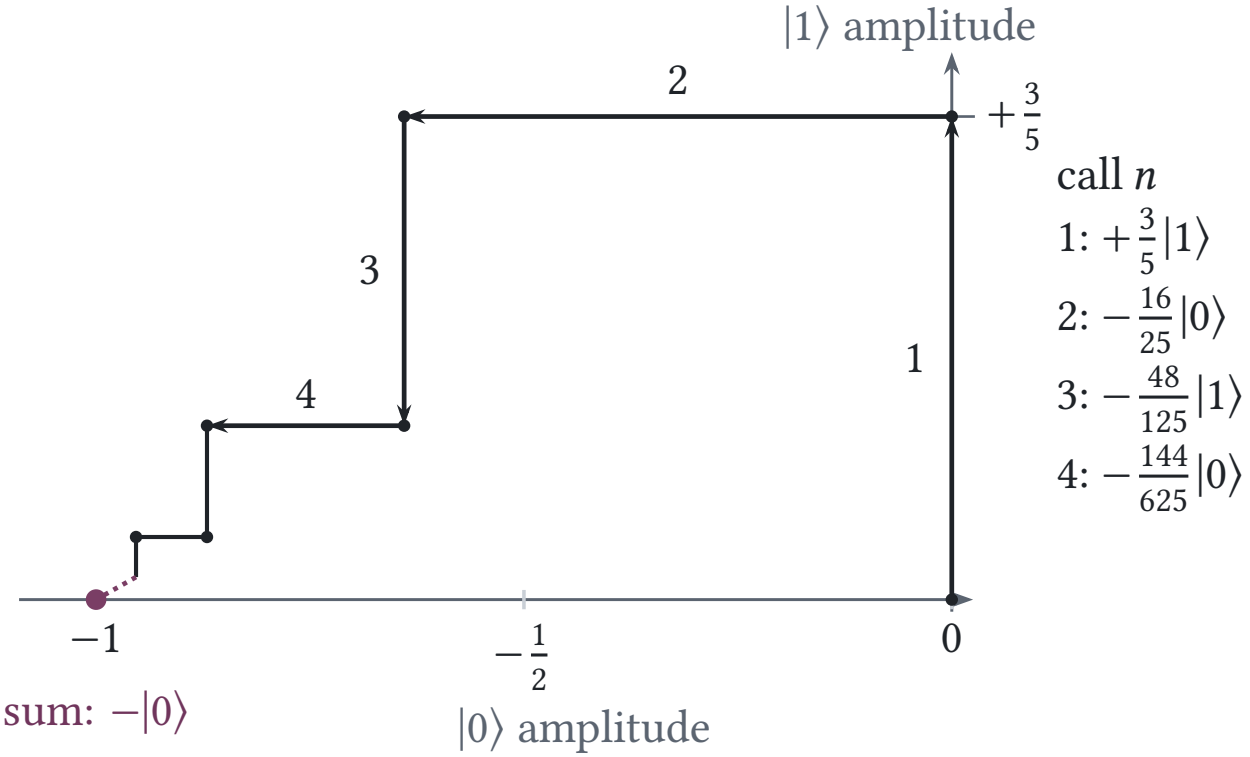
(a) `trace`*: add the exit vectors*



call n

1: $+\frac{3}{5}|1\rangle$

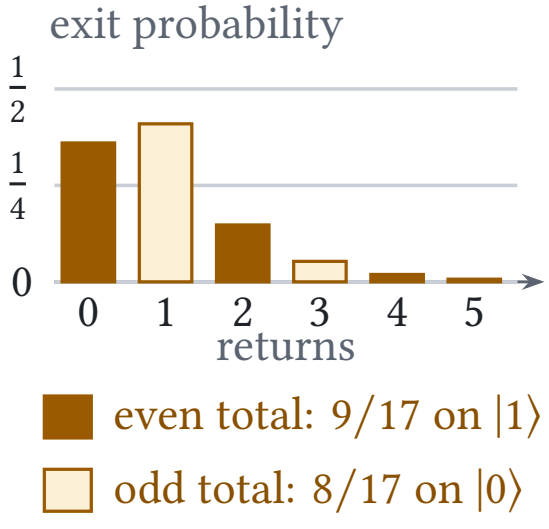
2: $-\frac{16}{25}|0\rangle$

3: $-\frac{48}{125}|1\rangle$

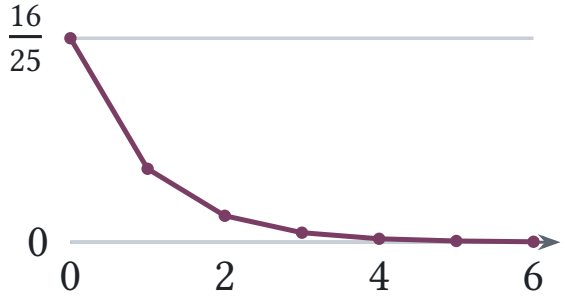
4: $-\frac{144}{625}|0\rangle$

each arrow adds one exit
intermediate points are unnormalized sums

(b) `while`*: add exit probabilities*



(c) probability still inside



returns k (after $k+1$ calls)
each further round retains 9/25